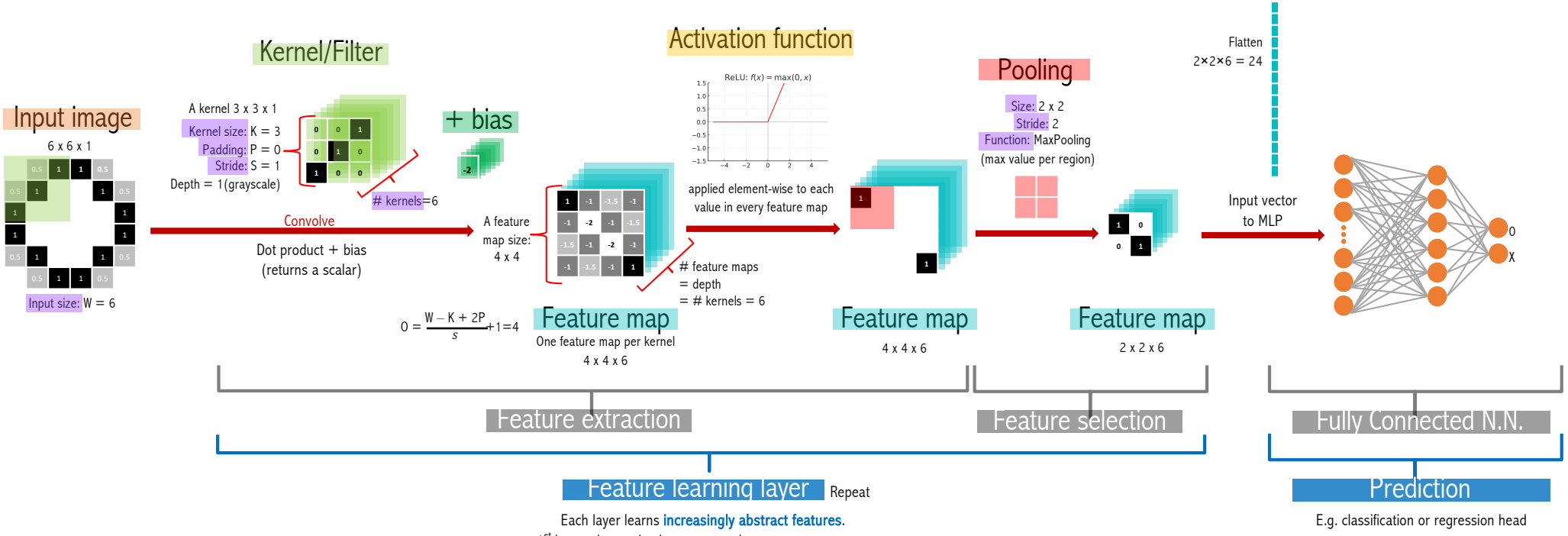
Convolutional Neural Network (CNN)

Hyperparameter choices

Unlike a simple MLP, which ignores spatial structure, CNNs exploit the local connectivity of images. Convolutions detect local patterns (edges, corners, textures), pooling reduces sensitivity to small spatial shifts, and stacking layers builds hierarchical features (from edges → shapes → objects). This makes CNNs far more efficient and accurate for image data than dense MLPs.



Kernel/Filter

- Kernel depth = input depth. E.g., for first layer grayscale: K x K x 1, for first layer RGB: K x K x 3; after (in img. above), 6 feature maps: K x K x 6
- Convolution = dot product + bias → scalar output (1 pixel)
- We define the size of the kernel, but the values inside the kernel (the weights) and biases are learned during training.
- We usually apply many kernels in parallel → multiple feature maps
- Each kernel detect the same feature but in different location. Dot product measures "alignment" between vectors (0 if orthogonal). So, the convolution operator measures how well the filter fits the respective part of the image.
- Stride: Controls how far the kernel shifts.

Feature map

- Represents the presence of a feature in different spatial locations
- Several features (filters) → several feature maps (stacked)
- Padding: avoids shrinking and helps preserve information at borders. Controls output size.
- Depth of feature maps = number of kernels applied

Pooling

- Reduces spatial size → fewer parameters → less computation
- Provides invariance to small translations/shifts
- Increases the reception field (sees more of the original image)
- Helps prevent overfitting and improves generalization
- Common: MaxPooling (selects strongest response)
- Usually, stride = pooling window

Made by: Laura Jimena Tagle M
v.1.0